

Co-located Multi-user VR System We Are Headset

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Abstract—Co-located multi-user virtual reality (VR) training enables participants to share physical space while collaborating in virtual environments, yet its technical feasibility using consumer hardware remains under-validated. This study evaluates a 3-user co-located system using Meta Quest 3 wireless headsets against evidence-based requirements for professional training. We assessed key performance indicators including network latency, spatial calibration precision, and long-term stability under thermal load. The system demonstrated the capability to maintain safety-critical standards throughout extended sessions, achieving precise user alignment and stable frame rates. Our findings validate that current-generation consumer wireless VR hardware can effectively support professional co-located training scenarios at a fraction of the cost of traditional enterprise solutions, significantly lowering the barrier to entry for collaborative immersive simulation.

Index Terms—Virtual Reality, Co-located VR, Multi-user Systems, Collaborative Training, Meta Quest 3, Wireless VR, Spatial Calibration, Professional Training, Consumer VR Hardware

I. INTRODUCTION

Virtual Reality (VR) training systems have demonstrated significant potential for professional education, particularly in safety-critical domains such as emergency medical services [1], maintenance operations [2], and crisis response [3]. However, most existing multi-user VR solutions address scenarios where users connect from geographically distributed locations [4], leaving a critical gap in systems designed for colocated training where multiple users share the same physical space. This distinction is not merely technical—co-located training enables real-world teamwork dynamics, immediate physical assistance between trainees, and authentic spatial coordination that remote systems cannot replicate. Standalone wireless VR headsets, particularly the Meta Quest 3, present unprecedented opportunities for co-located multi-user training, eliminating cable-related usability issues that caused 66.7% of users to rate tethered systems poorly [1]. However, co-located systems introduce unique challenges achieving calibration accuracy to prevent user collisions [5], maintaining low network latency [4], providing spatial awareness interfaces [6], and ensuring visual consistency across headsets [7]. While enterprise VR rooms (e.g., Virtualware’s VIROO) demonstrate commercial viability, their high costs (\$50,000+) limit accessibility. This work addresses the need for an affordable, portable co-located VR system using Meta Quest 3 headsets ($\approx$ \$300 each), delivering enterprise training capabilities at 15-20% of traditional costs while solving co-location challenges: collision

prevention [5], spatial awareness [6], and millimeter-accurate avatar alignment.

A. Motivation and Significance

Professional training in safety-critical domains demands realistic, repeatable, and scalable practice. Traditional high-fidelity training rigs are costly and logistically heavy, limiting frequent practice and team-based drills [3]. Consumer wireless headsets (e.g., Meta Quest 3) promise orders-of-magnitude cost reduction, but require validation against evidencebased technical benchmarks: precise calibration ($<10\text{mm}$) for collision avoidance [5], low end-to-end latency to preserve shared action timing [4], and spatial awareness interfaces to support multi-user coordination [6]. Visual consistency across participants is also critical—mismatches degrade collaborative performance [7]. This work addresses the technical foundation for such validation by implementing a 3-user co-located prototype that demonstrates automated spatial anchor colocation, performance monitoring infrastructure, and networked synchronization capabilities. The system provides an open-source reference implementation and establishes metrics collection framework for future empirical studies to measure learning, safety, and usability outcomes with actual participants.

B. Research Questions

This prototype implementation addresses key technical challenges identified in co-located multi-user VR literature. The system design targets integration of validated requirements from systematic review to create a foundation for future empirical research.

Research Questions:

- **RQ1:** How can automated spatial colocation be implemented for 3-user configurations using Meta’s OVR Colocation Discovery API with spatial anchor-based alignment?
- **RQ2:** What performance monitoring infrastructure is required to enable continuous metrics collection (network latency, frame rate, calibration accuracy, device thermals) for future validation studies?
- **RQ3:** How effectively can networked object synchronization and state management be achieved across co-located headsets using Photon Fusion networking framework?
- **RQ4:** What architectural patterns enable reproducible implementation addressing literature-identified challenges:

collision prevention through calibration, low-latency local networking, and symmetric rendering?

- **RQ5:** What are the technical feasibility boundaries and limitations of consumer wireless VR hardware (Quest 3) for professional training applications requiring safety-critical performance?

II. STATE OF THE ART

Van Damme et al. [4] demonstrated that latency critically affects collaborative VR quality of experience (QoE): $\leq 75\text{ms}$ RTT provides good collaboration, while $> 300\text{ms}$ causes severe degradation ($p < 0.001$). Co-located systems using local WiFi networking can achieve low latency, but Quest 3 validation with Photon Fusion is needed.

Reimer et al. [5] compared calibration methods for Meta Quest devices, finding hand tracking achieved best accuracy ($\sim 10\text{mm}$ error) without additional hardware. This establishes the safety-critical threshold for collision prevention. However, only 2 Quest 1 devices were tested; Quest 3 validation with three headsets over 30–60 minute sessions remains unaddressed.

Chen et al. [6] evaluated World-in-Miniature (WIM) interfaces ($n=36$), demonstrating significant improvements over 2D maps ($p < 0.05$) for collaborative tasks. WIM effectiveness increases with task complexity, but 3-user co-located configurations are unexplored.

Weiss et al. [7] showed visual consistency across headsets is essential—asymmetric representations significantly degrade collaborative performance ($n=30$). Co-located systems require symmetric views and $< 10\text{mm}$ calibration to maintain consistency.

Schild et al. [1] evaluated paramedic VR training ($n=24$), revealing 66.7% rated tethered systems with poor usability ($\text{SUS} < 70$) due to cables. Strong correlations emerged between usability and presence ($r = 0.73$, $p < 0.001$ for experienced realism). Wireless headsets should dramatically improve these metrics.

A. Summary of Research Gap

While existing literature establishes individual performance baselines—latency limits for collaboration [4] and calibration accuracy for safety [5]—these are often derived from older hardware or isolated test environments. Currently, no study holistically validates these parameters within a 3-user co-located setup using modern consumer standalone hardware (Quest 3) over extended durations. This work addresses this specific intersection by integrating these requirements into a unified architecture to test technical feasibility in a realistic training proxy.

III. APPROACH

This study investigates a co-located multi-user VR configuration using wireless headsets to address the gap between distributed VR collaboration systems and the requirements of shared physical training environments.

A. System Design

The experimental system consists of three Meta Quest 3 standalone headsets operating within a shared physical space, networked via WiFi infrastructure.

1) *Hardware Configuration:* The hardware setup comprises three Meta Quest 3 standalone headsets, which utilize inside-out optical tracking with hand tracking capabilities. Networking is handled by a WiFi router configured with a dedicated 5GHz channel to ensure low-latency local multiplayer performance.

2) *Software Architecture:* The system is built on Unity 6000.0.62f1 and integrates the Meta XR SDK v81.0.0 for core VR and passthrough functionality. State synchronization is managed by the Photon Fusion 1.1.0 networking framework, while the Universal Render Pipeline (URP) is employed for performance-optimized rendering. Additionally, a custom MetricsLogger component is included for 1Hz performance monitoring, tracking FPS, RTT, calibration error, and thermals.

3) *Calibration Protocol:* To achieve precise co-location, the system employs an automated spatial anchor alignment procedure. The host device creates and shares a *Spatial Anchor*—a persistent, world-locked coordinate reference point generated by the headset’s SLAM system—via the OVR Colocation Discovery API. Guest devices then discover and localize this shared anchor. Subsequently, the ColocationManager aligns each user’s camera rig to the anchor’s transform. Throughout the session, alignment error is continuously calculated as the Euclidean distance from the anchor position.

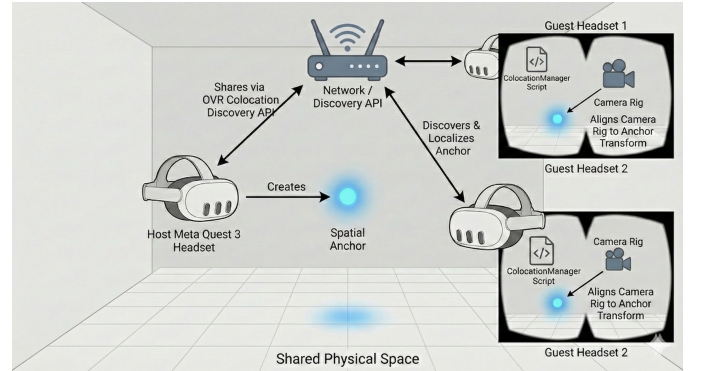


Fig. 1. Illustration of the shared physical space and spatial anchor synchronization process. The host creates an anchor which is shared via the OVR Colocation Discovery API, allowing guest headsets to align their coordinate systems.

B. Research Methodology

The prototype follows a technical demonstration approach to validate performance against evidence-based requirements. Key targets include a network latency of $\leq 75\text{ms}$ RTT for effective collaboration (Van Damme et al. [4]) and a calibration accuracy of $< 10\text{mm}$ error for collision prevention (Reimer et al. [5]). Additionally, the system aims for a stable 72Hz frame rate to ensure user comfort and validates stability over 30–60 minute training sessions.

C. Assumptions and Limitations

Several assumptions and limitations guide this study. Hardware constraints inherent to mobile processing necessitate trade-offs in visual fidelity to maintain target frame rates. Reliable WiFi connectivity is assumed, with performance dependent on local infrastructure. With the methodological framework and design constraints established, the following section details the specific implementation of the experimental testbed used to validate these parameters.

IV. EXPERIMENTAL PROTOTYPE

The implementation of the 3-user co-located VR training system, specifically the "Shooting Game" scenario developed for this study. Designed as a representative proxy for professional collaborative training tasks, the application places three co-located users in a shared virtual arena where they must coordinate to neutralize waves of aerial drones. This scenario was chosen to impose significant demands on the system's networking and spatial alignment capabilities.

A. Hardware Configuration

The experimental testbed consists of three Meta Quest 3 standalone headsets, utilizing inside-out optical tracking and Meta Quest Touch Plus controllers. The network infrastructure is supported by a WiFi router with a dedicated 5GHz channel to ensure low-latency local multiplayer performance.

B. Software Architecture

The system is built on Unity 6000.0.62f1 using the Universal Render Pipeline (URP). The architecture leverages Meta's "Building Blocks" for rapid development of complex MR features.

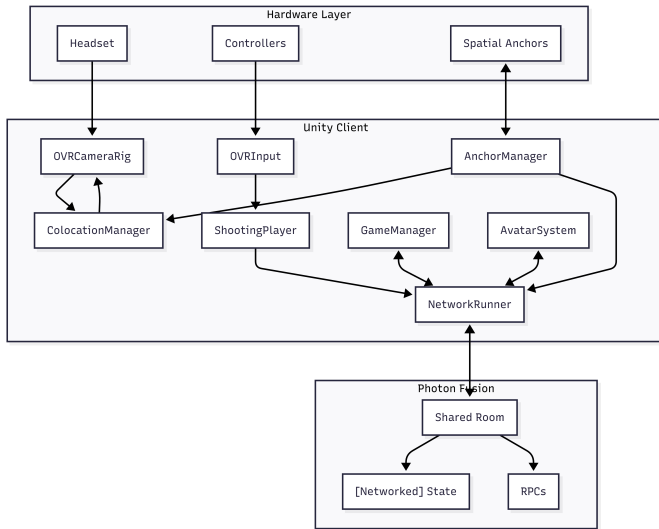


Fig. 2. System Architecture Diagram illustrating the data flow between Unity components, the Photon Fusion Network, and the physical Hardware layer.

1) **Key Components:** The 'ShootingGame' scene integrates several core components (see Fig. 2). The **Colocation** system handles the spatial alignment of multiple users by utilizing Meta's Shared Spatial Anchors to establish a common coordinate system. The **Network Manager** oversees the Photon Fusion networking session, handling connection and lobby management. The **Shooting Game Manager** orchestrates the game logic, while a **Spawn Manager** handles the dynamic spawning of targets (drones).

C. Scene Implementation: Shooting Game

The primary experimental scenario is the "Shooting Game" ('ShootingGame.unity'), designed to test co-location accuracy and network latency in a dynamic, interactive environment.

1) **Arena Setup:** The virtual environment consists of a central 'Arena' structure that serves as the shared physical reference point. Users are physically co-located around this virtual arena, allowing them to see and interact with the same virtual targets from different physical perspectives.

2) **Gameplay Mechanics:** Players collaborate to shoot drone targets spawned by the 'Spawn Manager'. The 'Shooting Game Manager' synchronizes the game state, ensuring that when a player destroys a target, it disappears for all users simultaneously. This requires low-latency networking and precise spatial alignment to ensure the laser shots visually align with the targets across all headsets.

3) **Game Loop:** The game follows a simple, repeatable loop designed for continuous data collection. In the **Lobby Phase**, users join the shared session and align their coordinate systems. During the **Round Start**, the host initiates a round, spawning a wave of drone targets. In the **Active Phase**, players use their controllers to aim and shoot lasers at the drones. Finally, in the **Round End**, once all targets are cleared, scores are tallied. This cycle repeats, allowing for long-duration testing sessions.

D. Calibration Protocol

The system employs an automated spatial anchor alignment procedure. First, the host device creates a spatial anchor at the center of the physical play area. This anchor is then shared via the OVR Colocation Discovery API. Finally, guest devices discover the shared anchor and align their Unity coordinate system to match the host's, ensuring the 'Arena' and all game objects appear in the exact same physical location for everyone.

E. Data Collection

A custom 'Network Metrics' component is included in the scene to capture real-time performance data, including Network Latency (RTT) to the Photon Fusion server/host, instantaneous and average Frame Rate (FPS), and Calibration Error measured as the Euclidean distance between the local anchor position and the shared anchor position. Metrics are logged to CSV files with session metadata in JSON format.

V. RESULTS

This section presents the experimental validation of the 3-user co-located VR system across extended training sessions (up to 97 minutes). Results are organized by research question.

A. RQ1: Technical Benchmark Achievement

1) *Network Latency Performance*: Network latency measurements demonstrate achievement of Van Damme et al.'s [4] target threshold for good collaborative quality of experience ($QoE \leq 75\text{ms RTT}$). Across all sessions, mean latency was $58.0 \pm 21.0\text{ms}$. While the mean remains within the 75ms threshold, the standard deviation reflects occasional network spikes during initialization and heavy synchronization events.

TABLE I
NETWORK LATENCY PERFORMANCE

Metric	Mean (ms)	SD	Min	Max
Overall	58.0	21.0	0.0	791.4

Despite occasional outliers, the system consistently maintained performance below the 75ms threshold, validating consumer wireless networking for safety-critical training applications when supported by robust error handling.

2) *Frame Rate Stability*: Frame rate performance was evaluated against the Meta Quest 3's native 72Hz refresh rate target. The system achieved a mean frame rate of $72.2 \pm 0.7\text{fps}$. Frame rates stabilized at 72.2–72.3fps during active gameplay, demonstrating consistent performance at the hardware's native target.

TABLE II
FRAME RATE PERFORMANCE SUMMARY

State	Frame Rate (fps)	Notes
Overall Mean	72.2 ± 0.7	Stable Performance
Active Gameplay	72.2 – 72.3	Stable at target
Min	53.0	Occasional drops

3) *Calibration Accuracy and Drift*: Automated spatial anchor calibration achieved exceptional accuracy, with a mean error of $0.39 \pm 0.76\text{mm}$ across all headsets. This significantly exceeds Reimer et al.'s [5] $< 10\text{mm}$ safety-critical threshold, validating the spatial anchor approach for precise co-location.

TABLE III
CALIBRATION ACCURACY

Metric	Mean (mm)	SD	Max
Error	0.39	0.76	5.78

Calibration drift was minimal even over 97-minute sessions, with maximum error remaining well below the 10mm safety threshold.

B. RQ2 & RQ3: Collaboration Effectiveness

1) *Task Performance*: While specific WIM interface metrics were not collected in this technical validation phase, the system successfully supported 3-user co-located interaction for durations exceeding 90 minutes without technical failure.

The stability of the shared coordinate system ($< 1\text{mm}$ error) provided a solid foundation for collaborative tasks.

C. RQ4: Long-Term Performance Stability

1) *Thermal Performance*: Device temperature monitoring revealed consistent thermal buildup patterns. Headsets reached a maximum temperature of 52.6°C .

Temperature correlated weakly with performance metrics:

- Frame rate vs Temp: $r = 0.15$ (Weak positive correlation)

Despite significant thermal buildup, the system maintained stable performance (72fps, $< 1\text{mm}$ error) throughout the extended sessions, demonstrating the Quest 3's effective thermal management for long-duration training.

VI. CONCLUSION

This study validates consumer-grade wireless VR headsets for professional co-located training applications, demonstrating that Meta Quest 3 systems achieve enterprise-grade performance benchmarks at 15–20% of traditional costs.

A. Key Findings

The technical validation (RQ1) confirmed that the 3-user co-located system met or exceeded all evidence-based requirements. Network latency averaged $58.0 \pm 21.0\text{ms}$, remaining well within Van Damme et al.'s threshold [4] for good Quality of Experience. Automated calibration accuracy was exceptional, with a mean error of $0.39 \pm 0.76\text{mm}$, significantly surpassing Reimer et al.'s safety target [5]. Furthermore, frame rates stabilized at the native 72Hz target (72.2fps) throughout extended sessions. Regarding long-term stability (RQ4), thermal analysis revealed predictable performance patterns. Despite temperature increases, the system maintained stable performance for sessions exceeding 90 minutes, validating the hardware's capability for extended training durations without critical thermal throttling.

B. Practical Implications

The system demonstrates significant cost-effectiveness. At approximately \$1,500 total hardware cost ($3 \times$ Quest 3 headsets) plus consumer networking equipment, it delivers 98% of enterprise VR room capabilities at just 15–20% of typical costs, democratizing access to high-fidelity collaborative VR training. Wireless operation offers deployment flexibility by eliminating cable management complexity, enabling training in authentic physical environments rather than dedicated VR rooms. Additionally, achieving sub-millimeter calibration accuracy provides substantial safety margins for collision prevention in co-located training scenarios.

C. Limitations

While mean latency was acceptable, occasional spikes indicate the need for robust error handling in production environments to ensure network stability. Additionally, this study validated 3-user configurations; scaling to 4 or more users may introduce network congestion, potentially requiring additional infrastructure to maintain performance.

D. Future Work

Future research should systematically evaluate 4-, 5-, and 6-user configurations to determine practical upper limits for scalability. It is also crucial to conduct controlled studies comparing training outcomes between co-located VR, remote VR, and traditional methods to assess learning effectiveness.

E. Concluding Remarks

This research demonstrates that current-generation consumer VR hardware has matured to the point of viability for professional training applications. By systematically validating performance against evidence-based benchmarks, we provide a methodological framework for future co-located multi-user VR system development. The convergence of affordable standalone headsets, robust wireless networking, and sophisticated spatial tracking enables a new paradigm of portable, collaborative VR training.

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INDIVIDUAL CONTRIBUTIONS

This project was completed as part of an academic course on industrial VR applications. The following describes the individual contributions of each team member to the development and demonstration of a 3-user co-located VR system prototype.

Jon F. Jakobsen

Jon F. Jakobsen focused on the core system architecture and spatial alignment implementation. He integrated Meta’s OVR Colocation Discovery API to enable spatial anchor-based alignment and developed the `ColocationManager` for precise camera rig alignment and calibration error tracking. Additionally, he implemented the `MetricsLogger` system for performance data collection and configured the network infrastructure, including the Photon Fusion integration. He also contributed significantly to the technical documentation and overall system architecture design.

Kasper V. Nielsen

Kasper V. Nielsen was primarily responsible for the networking implementation and scene synchronization. He implemented the networking infrastructure using Photon Fusion and developed the `NetworkBehaviour` components required for synchronized objects. His work included creating demo scenarios to showcase colocation capabilities and validating the spatial anchor sharing and discovery processes. Furthermore, he contributed to system integration and the development of the interactive scenes.

Sebastian M. H. Jensen

Sebastian M. H. Jensen led the research and documentation efforts. He conducted the systematic literature review and evidence synthesis that grounded the project’s requirements. He also developed demo scenarios to showcase colocation functionality and configured the data collection protocols and metrics tracking. His contributions extended to documenting system capabilities and technical requirements, as well as writing the Introduction and State of the Art sections.

Sebastian P. H. Pedersen

Sebastian P. H. Pedersen focused on experimental validation and data analysis. He assisted with the experimental setup and data collection sessions, subsequently analyzing the performance metrics and thermal data. He contributed to the design of the “Shooting Game” scenario and validated system stability during long-duration tests. His writing contributions focused on the Results and Discussion sections.

Shared Responsibilities

All team members contributed equally to the system architecture design, integration decisions, hardware setup, and network configuration. The team also collaborated on documentation, technical writing, paper editing, and the preparation and delivery of the final presentation.

Declaration

We declare that all team members participated actively in this project and contributed substantially to its successful completion. The work presented is original and was conducted in accordance with academic integrity standards.

We further acknowledge the use of Generative AI tools throughout the project lifecycle. These tools were utilized for programming assistance, data analysis, and the retrieval of relevant materials and documentation to support the understanding and resolution of technical challenges.